

### 3.2.1 Exercise physiology

Students should understand the adaptations to the body systems through training or lifestyle, and how these changes affect the efficiency of those systems.

#### 3.2.1.1 Diet and nutrition and their effect on physical activity and performance

| Content                                                                             | Additional information                                                                                                                                                                                               |
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| Understand the exercise-related function of food classes.                           | Carbohydrate.<br>Fibre.<br>Fat (saturated fat, trans fat and cholesterol), protein, vitamins (C,D, B-12, B-complex), minerals (sodium, iron, calcium), water (hydration before, during and after physical activity). |
| Positive and negative effects of dietary supplements/manipulation on the performer. | Creatine, sodium bicarbonate, caffeine, Glycogen loading.                                                                                                                                                            |

#### 3.2.1.2 Preparation and training methods in relation to maintaining physical activity and performance

Students should understand quantitative methods, the types and use of data for planning, monitoring and evaluating physical training, and to optimise performance.

| Content                                                                           | Additional information                                                                                                   |
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| Understanding of the key terms relating to laboratory conditions and field tests. | Quantitative and qualitative.<br>Objective and subjective.<br>Validity and reliability.                                  |
| Physiological effects and benefits of a warm-up and cool down.                    | Stretching for different types of physical activity (static and ballistic).                                              |
| Principles of training.                                                           | Specificity, progressive overload, reversibility, recovery, Frequency Intensity Time Type of Training (FITT) principles. |
| Application of principles of periodisation.                                       | Macro cycle, Meso cycle, Micro cycle.<br>Preparation, competition, transition.<br>Tapering, peaking.                     |

| Content                                                  | Additional information                                                                                                                                                                                                                                                              |
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| Training methods to improve physical fitness and health. | <p>HIIT/interval training (anaerobic power).</p> <p>Continuous training (aerobic power).</p> <p>Fartlek (aerobic power).</p> <p>Circuit training (muscular endurance).</p> <p>Weight training (strength).</p> <p>Proprioceptive Neuromuscular Facilitation (PNF) (flexibility).</p> |

### 3.2.1.3 Injury prevention and the rehabilitation of injury

| Content                                                                                 | Additional information                                                                                                                                                                                                                                                                                                                                                                                                          |
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| Types of injury.                                                                        | <p>Acute (fractures, dislocations, strains, sprains).</p> <p>Chronic (achilles tendonitis, stress fracture, 'tennis elbow').</p>                                                                                                                                                                                                                                                                                                |
| Understanding different methods used in injury prevention, rehabilitation and recovery. | <p>Injury prevention methods: Screening.</p> <p>Protective equipment.</p> <p>Warm up, flexibility training (active, passive, static and ballistic), taping and bracing.</p> <p>Injury rehabilitation methods (proprioceptive training, strength training, hyperbaric chambers, cryotherapy, hydrotherapy).</p> <p>Recovery from exercise (compression garments, massage/foam rollers, cold therapy, ice bath, cryotherapy).</p> |
| Physiological reasons for methods used in injury rehabilitation.                        | Hyperbaric chambers, cryotherapy.                                                                                                                                                                                                                                                                                                                                                                                               |
| Importance of sleep and nutrition for improved recovery.                                |                                                                                                                                                                                                                                                                                                                                                                                                                                 |